

Soil Dataset Documentation

soil_classifier project

Dataset 1 - Soil Image Dataset (Kaggle)

Source:	kaggle.com/datasets/jayaprakashpondy/soil-image-dataset
License:	Unknown - verify before commercial use
Size:	1,553 images, ~144 MB
Classes:	Alluvial, Black, Clay, Red soil
Split:	Pre-split Train / Test folders included
Year:	Not specified
Location:	India (dominant Indian agricultural soil types)
Camera angle:	Ground-level, ~0.3-1m above soil surface, downward facing

Close-up smartphone photos of soil surfaces under varied lighting. The four classes reflect the main Indian agricultural soil classifications. Widely cited in published ML research including AgroSense and SoilNet.

Dataset 2 - Soil Moisture Dataset (Mendeley Data)

Source:	data.mendeley.com/datasets/skcc44yvvg/2
License:	CC BY 4.0 - free for all uses including commercial
Size:	1,177 images (Before Augmentation), 3 classes
Classes:	Dry, Moderate, Wet
Split:	No pre-existing split - 80/20 applied at training time
Camera:	Sony Xperia 1 Mark II smartphone
Camera angle:	Ground-level, natural outdoor lighting
Location:	Not specified
Year:	Not specified, dataset version 2

Explicitly captured with a consumer smartphone making it the most realistic proxy for phone-app deployment. Class distribution is balanced: Dry (387), Moderate (384), Wet (406). An augmented version (~11,000 images) was excluded to prevent validation data leakage.

Training Notes

Active model: soil type classifier on Dataset 1 (4 classes, EfficientNet-B0).

Dataset 2 retained for a future soil condition model or multi-output model.